THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Assertion (A): A straight current-carrying wire produces a magnetic field around it. Reason (R): A compass needle placed near the wire gets deflected when the current flows. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Decomposers are called the cleaners of the environment. Reason (R): They break down dead organisms into simple substances. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Mg burns in oxygen: 24 g Mg gives 40 g MgO. Mass of oxygen used is: 1

(a) 64 g **(b)** 40 g **(c)** 24 g **(d)** 16 g

4. 20 mL of NaOH is neutralised by 10 mL of HCl. To neutralise 50 mL of the same NaOH, the volume of HCl needed is: 1

(a) 5 mL **(b)** 25 mL **(c)** 10 mL **(d)** 100 mL

5. Four metals P, Q, R, S: P displaces Q from its salt; Q displaces R; R displaces S. The least reactive is: 1

(a) Q **(b)** R **(c)** P **(d)** S

6. In the balanced equation $2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$, the coefficient of water is: 1

(a) 6 **(b)** 7 **(c)** 3 **(d)** 4

7. In photosynthesis, six molecules of carbon dioxide combine with water to form how many molecules of glucose? 1

(a) 6 **(b)** 1 **(c)** 2 **(d)** 12

8. A student touches a hot object, withdraws the hand and then feels the heat. The reason is: 1

(a) motor neuron is slower
(b) no relay neuron
(c) brain acts before spinal cord
(d) reflex action is faster than brain sensation

9. Rhizopus is grown on bread in a sealed, completely dry, sterile packet. Growth is absent because 1

(a) spores are not produced **(b)** spores need moisture to germinate
(c) bread kills them **(d)** sporangia are absent

10. In a cross Tt x Tt, three offspring are tall. The probability that the fourth is dwarf is 1

(a) 1/4 **(b)** 1/2 **(c)** 1 **(d)** 3/4

11. A concave mirror forms a virtual image of magnification +2. The object lies at 20 cm. The focal length is 1

(a) -40 cm **(b)** -20 cm **(c)** -10 cm **(d)** -80 cm

12. A person can see only from 40 cm to 100 cm. The lens power for distance and reading lens power (to read at 25 cm) are respectively 1

(a) -0.4 D and +3 D **(b)** -1 D and +1.5 D
(c) -1 D and +1 D **(d)** -2.5 D and +1.5 D

13. A wire is stretched to three times its length with its volume unchanged. The new resistance is 1

(a) R / 3 **(b)** R / 9 **(c)** 3 R **(d)** 9 R

14. A coil moves along the field lines of a uniform field in a direction parallel to the field. The induced current is 1

(a) alternating **(b)** zero **(c)** maximum **(d)** half

15. If the energy available at the producer level is 4000 kJ, the energy transferred to the third trophic level is 1

(a) 4000 kJ **(b)** 400 kJ **(c)** 4 kJ **(d)** 40 kJ

16. A student places metal M in blue copper sulphate solution. The blue colour fades to colourless and a reddish-brown solid forms. M could be: 1

(a) mercury **(b)** gold **(c)** silver **(d)** zinc

17. A salt X, when heated, loses water and turns from blue to white. X is: 1

(a) $CuSO_4 \cdot 5\,H_2O$ **(b)** $Na_2CO_3 \cdot 10\,H_2O$ **(c)** $FeSO_4 \cdot 7\,H_2O$ **(d)** $CaSO_4 \cdot 2\,H_2O$

18. Green copper carbonate on heating gives black solid X. X, when heated in hydrogen, gives a brown metal. The metal is: 1

(a) lead **(b)** iron **(c)** copper **(d)** zinc

19. An organic compound Y gives effervescence with $NaHCO_3$ and is used as a preservative. Y is: 1

(a) ethanol **(b)** methanal **(c)** ethanoic acid **(d)** ethane

20. A leaf variegated with green and white patches shows starch only in green regions because: 1

(a) white parts are dead **(b)** white parts get no light

(c) chlorophyll is needed for photosynthesis **(d)** starch is stored there

21. A woman has swollen neck and slow metabolism. Her diet is deficient in: 1

(a) iron **(b)** iodine **(c)** zinc **(d)** calcium

22. If the placenta fails to form properly, the main problem for the embryo is 1

(a) excess heat **(b)** shortage of nutrients and oxygen

(c) lack of chromosomes **(d)** lack of DNA

23. In pea plants F2 seeds show round-green and wrinkled-yellow types that were absent in both parents. This happens due to 1

(a) new combinations of independently inherited genes

(b) loss of genes in F1

(c) acquired traits

(d) blending of traits

24. When the object distance in a convex lens is made 3 times f, the image distance is 1

(a) 0.75 f **(b)** 2 f **(c)** 1.5 f **(d)** 3 f

25. The sky appears blue because molecules of the atmosphere scatter 1

(a) only yellow light **(b)** blue light more strongly than red light

(c) all colours equally **(d)** red light more strongly than blue light

26. In the circuit, ammeter reads 2 A and voltmeter across the resistor reads 10 V. The resistance is 1

(a) 0.2 ohm **(b)** 5 ohm **(c)** 20 ohm **(d)** 12 ohm

27. A magnet is brought toward a coil at a speed v and the induced emf is E. If the speed is tripled, the emf becomes 1

(a) E **(b)** 3 E **(c)** 9 E **(d)** E / 3

28. A student claims that eating at a lower trophic level gives more energy. This is because 1

(a) food at lower levels is tastier **(b)** the Sun is nearer

(c) plants are free **(d)** lower levels have more energy

29. Assertion (A): Chips packets are flushed with nitrogen. Reason (R): Nitrogen is a reactive gas that speeds up oxidation of oil. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

30. Assertion (A): Aqueous solutions of glucose and alcohol do not conduct electricity. Reason (R): They do not ionise in water to give H^+ ions. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

31. Assertion (A): Ionic compounds are brittle. Reason (R): In ionic crystals, like ions can come near each other on a slight shift, causing repulsion. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

32. Assertion (A): Diamond is a poor conductor of electricity but graphite conducts. Reason (R): In graphite, each carbon is bonded to four others. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

33. Assertion (A): Bile juice is necessary for digestion of fats. Reason (R): Bile contains the enzyme lipase that digests fats. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

34. Assertion (A): Plant roots show positive geotropism. Reason (R): Roots grow upwards against gravity. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

35. Assertion (A): Testes of mammals lie outside the abdominal cavity. Reason (R): Sperm formation requires a lower temperature than the body temperature. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

36. Assertion (A): A recessive trait is not seen in the F1 generation. Reason (R): The dominant allele masks the recessive allele in heterozygotes. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

37. Assertion (A): The image formed by a plane mirror is real. Reason (R): Plane mirror forms an image behind the mirror. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

38. Assertion (A): Violet light is deviated most by a prism. Reason (R): The refractive index of glass is maximum for violet light. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

39. Assertion (A): The resistance of a conductor increases with temperature. Reason (R): At a high temperature, the electrons collide more often with the atoms. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

40. Assertion (A): Soft iron is used as the core of an electromagnet. Reason (R): Soft iron is easily magnetised and demagnetised. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

41. **Assertion (A):** An ammeter is connected in series in a circuit. 1
Reason (R): An ammeter has a very low resistance.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** Two magnetic field lines never intersect each other. 1
Reason (R): At any point in a magnetic field, the field has only one direction.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** Decomposers maintain the nutrient cycle in an ecosystem. 1
Reason (R): They break down dead organic matter into simple inorganic substances that return to the soil.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** A white precipitate is formed when barium chloride solution is mixed with sodium sulphate solution. 1
Reason (R): The reaction between barium chloride and sodium sulphate is a double displacement reaction.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** Bleaching powder is used for disinfecting drinking water. 1
Reason (R): Bleaching powder is produced by the action of chlorine on dry slaked lime.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** Aluminium oxide is an amphoteric oxide. 1
Reason (R): Aluminium is widely used for making cooking utensils.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** Ethanol is oxidised to ethanoic acid by alkaline potassium permanganate. 1
Reason (R): Acidified potassium dichromate is also an oxidising agent that converts ethanol to ethanoic acid.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** Arteries have thick, elastic walls. 1
Reason (R): Veins have valves that prevent backflow of blood.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** Auxin is synthesised at the shoot tip. 1
Reason (R): Gibberellins help in the growth of the stem.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** Hydra reproduces by budding. 1
Reason (R): Planaria can regenerate a complete individual from a cut piece.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**